

Kavindu Damsith

Computer Science & Engineering | Cloud, DevOps & Reliability

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Mawanella, Sri Lanka

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PROFILE

Computer Science & Engineering background with hands-on experience in CI/CD automation, containerised application delivery, Kubernetes configuration, workload scaling, monitoring, and runtime optimisation. Internship work included Jenkins pipelines, Docker image workflows, a multi-node Kubernetes environment, HPA/VPA policies, Prometheus monitoring, and JVM and pod-resource tuning. A strong software-development foundation across Java, Python, Node.js, Spring Boot, databases, and web applications supports practical troubleshooting from application code through deployment. Project work also covers Kustomize-based environments, OpenFeature and flagd experiments, multi-service healthcare deployment assets, and latency-sensitive video streaming.

EDUCATION

B.Sc. (Hons) in Computer Science & Engineering Requirements completed: Jun 2026
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level — Physical Science 2020
3 A passes

G.C.E. Ordinary Level 2017
9 A passes

CORE CAPABILITIES

CONTAINERS & ORCHESTRATION

Docker, Kubernetes, HPA, VPA, Kustomize, Harbor

CI/CD & AUTOMATION

Jenkins, Git, Bitbucket, Shell, Terraform, Ansible, Argo CD

CLOUD & OBSERVABILITY

Linux, AWS, Prometheus, JVM tuning, Resource limits, Autoscaling

DEVELOPMENT FOUNDATION

Java, Python, JavaScript, Spring Boot, Node.js, React, MySQL, MongoDB

PROFESSIONAL EXPERIENCE

Software Engineering Intern Dec 2024 – Jun 2025
Cloud Solutions International | Sri Lanka

- Supported application-delivery automation through Jenkins CI/CD pipelines and Docker image builds.
- Built a multi-node Kubernetes environment and configured horizontal and vertical pod autoscaling for containerised services.
- Tuned JVM settings and workload resources, using Prometheus monitoring to support performance investigation.
- Contributed testing and reliability improvements to Antikythera, an open-source Spring test-generation tool.

Freelance Developer 2020–2024
Fiverr | Remote

- Delivered machine-learning, automation, web-scraping, and embedded-prototyping solutions for clients across different domains.
- Worked directly with clients to implement, test, revise, and hand over solutions across different technical constraints.

Web Developer 2020–2024
FutureX

- Built, debugged, and maintained frontend and backend functionality in collaborative Git-based workflows.
- Built practical experience in application delivery, debugging, maintenance, and technical troubleshooting.

PROJECTS

Deployment engineering project

Kubernetes Scaling Challenge

Deployment environment for healthcare API services covering workload scaling, configuration separation, storage integration, and scripted rollouts.

- Defined Kubernetes deployments with horizontal and vertical pod autoscaling.
- Separated runtime configuration and sensitive values while integrating storage and CSI concerns.
- Created Kustomize and shell-based rollout steps for consistent environment setup.

Kubernetes | HPA | VPA | Kustomize | Shell | CSI

Private lab | Platform comparison

Feature Flag Platform Lab

Consistent Kubernetes environment for evaluating four open feature-flag platforms under comparable deployment conventions.

- Packaged FF4J, flagd/OpenFeature, Flagsmith, and Unleash using reusable Kustomize structures.
- Provisioned PostgreSQL-backed deployments and compared operational trade-offs with repeatable manifests.

Kubernetes | Kustomize | OpenFeature | PostgreSQL

Multi-service deployment project

HealthLink Deployment

Healthcare application combining web workflows, an MRI-analysis service, and Kubernetes deployment assets.

- Connected React, Node.js, MySQL, Flask, and ResUNet across the application workflow.
- Prepared Kubernetes deployment assets for the application services.

Kubernetes | React | Express | MySQL | Flask | ResUNet

[GitHub repository](#)

Open-source reliability contribution

Antikythera Test Generator

Contributed to Java tooling that analyses Spring controllers and dependencies to generate focused application test assets.

- Tested dependency-solver behaviour across repositories, interfaces, DTOs, annotations, and dependency graphs.
- Documented edge cases and strengthened JavaParser, Maven, JUnit, and Spring test coverage.

Java | Spring | JavaParser | Maven | JUnit

[GitHub repository](#)

Selected infrastructure work | CSI internship

Cloud Application Delivery & Kubernetes Operations

Built and configured a multi-node container environment while supporting healthcare-application delivery at Cloud Solutions International.

- Containerised services with Docker and automated delivery through Jenkins pipelines.
- Configured Kubernetes workloads, HPA/VPA policies, CPU and memory resources, and multi-node operation.
- Used Prometheus observations to support JVM and pod-resource tuning.

Linux | Jenkins | Docker | Kubernetes | HPA/VPA | Prometheus

Engineering experiment | Public repository

OpenFeature Load Lab

Spring Boot integration and concurrency experiment for understanding flagd-backed feature-flag evaluation under repeated access.

- Configured a TLS-connected flagd provider through the OpenFeature SDK.
- Built a reusable evaluation service and recorded behaviour across controlled batches of 4,000 lookups.

Java 17 | Spring Boot | OpenFeature | flagd | Maven

[GitHub repository](#)

Research engineering

VSR Streaming Console

Companion environment for a real-time video super-resolution research pipeline, combining media orchestration, GPU streaming, preview, and telemetry.

- Extracted source metadata through FFprobe and controlled native-FPS playback.
- Orchestrated CUDA/NVENC streaming, MJPEG preview, controls, and runtime metrics.

Python | Flask | FFmpeg | CUDA | React | PySide6

[GitHub repository](#)

Live commercial platform

Maple Wraps

Live full-stack platform for a Canadian event-floor-wrap business, developed across customer and administration surfaces.

- Built the React and TypeScript customer application, Next.js administration workspace, Express API, and MongoDB data layer.
- Implemented role-based access, configurable quotes, project workflows, dashboards, and a live storefront.

React | TypeScript | Next.js | Express | MongoDB

[Live platform](#)

ADDITIONAL PROJECTS

ForkNetRT

Real-time video super-resolution research balancing reconstruction quality, temporal consistency, inference cost, and latency.

Python | PyTorch | OpenCV | FFmpeg
[Research profile](#)

HyperTalk

Assistive communication platform translating sign language to speech and speech into an accessible visual workflow, recognised at the 2023 BrainStorm Innovation Competition.

Flutter | Django | TensorFlow | MediaPipe | OpenCV | TTS

Cinema Booking Platform

Transactional system with temporary seat reservations, secure payments and refunds, authentication, reviews, discounts, and real-time updates.

React | Express | Sequelize | MySQL | Stripe | Socket.IO

PlateShare

Multi-role food-sharing platform with mobile workflows, payments, maps, media, live order status, and relational persistence.

React Native | Express | Sequelize | MySQL | Stripe
[Repository](#)

Abacus Learning Platform

Commercial UK education platform with progressive lesson access, payments, notifications, file delivery, real-time updates, and cloud storage integration.

React | Node.js | MySQL | Socket.IO | Stripe | AWS S3

TrainGPS Safety Intelligence

Railway safety prototype combining ESP32 GPS capture, geospatial danger estimation, and real-time driver-attention monitoring.

Python | Jupyter | PHP | ESP32 | Geospatial ML
[Repository](#)

OPERATIONAL HIGHLIGHTS

Automated delivery

Jenkins pipelines, Docker image workflows, Kubernetes deployments, Git, Bitbucket, Harbor, and repeatable rollout steps.

Platform configuration

Kustomize, OpenFeature, flagd, PostgreSQL-backed services, configuration separation, and storage integration.

Reliability and scaling

HPA/VPA configuration, CPU and memory tuning, JVM optimisation, Prometheus monitoring, and workload investigation.

Application-aware operations

Hands-on troubleshooting across Java and Spring, Node.js, Python and Flask, MySQL, MongoDB, and React applications.

AWARDS & COMPETITIONS

1

MoraXtreme 9.0

First place, 2024, among more than 350 teams island-wide.

2

Hackdoze 1.0

First runner-up, 2021.

4

ENIGMA 24 - Crack the Code

Fourth place in the team finals, 2024.

1

CYPHER 2023 Hackathon

Winner, team competition, 2023.

3

BrainStorm Innovation Competition

Second runner-up, 2023, with the HyperTalk assistive-communication project.

8

MoraXtreme 8.0

Eighth place, 2023, among more than 300 teams.

REFEREES

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